

MODIFIED SIMPLE EXPRESSION FOR BANDGAP NARROWING IN *n*-TYPE GaAs

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Abstract—A simple theoretical expression which has been very successful in interpreting experiments on bandgap narrowing (BGN) in *n*- and *p*-type Si, *n*-type Ge and *p*-type GaAs is modified to fit recent optical and transistor BGN data for *n*-type GaAs. The impurity scattering part of the BGN in this expression is replaced by an expression obtained by a numerical fit with the results of multiple scattering theory. The modified expression is simple and describes all available luminescence and transistor data extremely well. The Fermi energy is evaluated by analyzing available luminescence data in *n*-type GaAs. It is found that for dopant concentrations larger than $2 \times 10^{19} \text{ cm}^{-3}$, the observed Fermi level starts becoming smaller than the calculated using the accepted value of conduction band density of states.

NOTATION

a	effective Bohr radius
BGN	bandgap narrowing
BGN _{imp}	bandgap narrowing due to impurity scattering
BGN _{mb}	bandgap narrowing due to many-body effects
ABGN	$= \Delta E_g^a$, apparent bandgap narrowing
ΔE^{FD}	Fermi–Dirac correction
D_p	diffusion coefficient of minority holes
E_F	Fermi level measured from the actual majority band edge of the doped crystal
ϵ	dielectric constant
A	correction factor
N_D	donor concentration
N_b	number of valleys in the conduction or valence band
n_{i0}	intrinsic carrier density
n_{ie}	effective intrinsic carrier density
R	effective Rydberg energy for majority band
$R_{(\text{mino})}$	effective Rydberg energy for minority band
r_a	equivalent to the average distance between impurity ions, given by

$$\frac{4\pi N_D r_a^3}{3} = 1$$

$$r_s \quad \text{many-body parameter, } = \frac{r_a}{a}$$

1. INTRODUCTION

BGN in Si[1] and in GaAs[2] is important for designing devices and interpreting experiments. Jain *et al.*[3] and Jain and Roulston[1] derived the following simple expression for BGN in all semiconductors:

$$\frac{\text{BGN}}{R} = 1.83 \frac{A}{r_s N_b^{1/3}} + \frac{0.95}{r_s^{3/4}} + \left[1 + \frac{R_{(\text{mino})}}{R} \right] \frac{1}{57} N_b r_s^{3/2}. \quad (1)$$

Symbols are defined in the Notation. A and N_b are known correction factors introduced to take peculiarities into account (i.e. anisotropy, multiple valleys and coupling between them, and distortion

due to heavy doping) of bands in individual semiconductors. If the bands are parabolic and isotropic, A and N_b become equal to unity and eqn (1) gives a universal curve for BGN in terms of the parameters R and r_s . Numerical values of the parameters (e.g. effective masses of carriers and dielectric constant needed to determine R and r_s and those of the correction factors for GaAs are given in the paper of Jain and Roulston. Note there are no fitting parameters in this equation. Jain *et al.*[1,2] have emphasized that classical theory of BGN and of band tails is incapable of describing the experimental results, and theories based on quantum mechanics are required. Though results of quantum mechanical theories have been highly successful in interpreting experiments, the theories are very complicated and in most cases results cannot be represented by simple expressions. Equation (1) is simple and yet all features of quantum mechanical theory are incorporated in it. Jain and Roulston[1] showed that the above expression agreed with all known luminescence and transistor data for BGN in *n*-Ge (no data are available on *p*-Ge), *n*- and *p*-Si and *p*-GaAs. More recently measurements of BGN have been extended to higher dopings in *p*-GaAs[4] and *n*-Si[5]. In both cases, experimental values of BGN agree with the predictions of Jain and Roulston's eqn (1).

It was noted by Jain and Roulston that the situation of *n*-GaAs was different. Though the BGN values of *n*-GaAs calculated using eqn (1) agreed very well with the more complete theory of Sernelius[6], they were considerably higher than the experimental values derived from luminescence by Yao and Compaan[7]. Recent independent measurements on transistors by Harmon *et al.*[8] have confirmed that the values of BGN in *n*-GaAs cannot be as high as predicted by this equation or by the more complete

theory of Sernelius[6]. Our attempt in this paper is the same as that of Jain and Roulston[1]; to obtain a closed-form simple expression for BGN in *n*-GaAs which is able to describe experiments.

2. MANY-BODY AND IMPURITY SCATTERING EFFECTS

Since the electron effective density of state mass in *n*-GaAs is small, BGN becomes large even at low doping densities. We will see in the next section that for $N_D = 4 \times 10^{17} \text{ cm}^{-3}$, BGN in *n*-GaAs is about 50 meV. In *n*- and *p*-Si, BGN is negligible at this doping concentration. At $N_D = 10^{18} \text{ cm}^{-3}$, BGN is about 30 meV in Si, 70–80 meV in *n*-GaAs. The lower doping range is therefore more important in *n*-GaAs. We should examine whether high-density theories are valid for *n*-GaAs, particularly in this range of doping.

Shift of band edges on heavily doping a semiconductor arises on two accounts: (1) many-body effects, i.e. the shifts due to exchange and correlation interactions; and (2) carrier-ionized impurity scattering. The condition for the validity of the many-body effects part of the high density theory, the first two terms on the right-hand side of eqn (1), is that $n_s/a < 1$. Since a is inversely proportional to the effective density of state mass, it is very large for *n*-GaAs. The range over which the many-body part of the calculation is valid is extended to the lower doping range. One expects that the calculation of this part by Sernelius[6], by Bennett *et al.*[9], by Abram *et al.*[10] and by using the first terms of eqn (1) should agree with each other. We will see in the next section that this indeed is the case.

The problem of impurity scattering is more complicated. Numerical calculations show the result of the last term of eqn (1) for impurity scattering agrees with Sernelius[6], who used proper RPA (random phase approximation) screening; the use of Thomas–Fermi screening in obtaining last term in eqn (1) cannot be responsible for this discrepancy. The approach used by Berggren *et al.*[11], by Sernelius[6], and by Jain and Roulston[1] in deriving eqn (1) neglects the effects of multiple scattering. The result of the high-density theory of Sernelius[6] and of Jain and Roulston[1] for impurity scattering in *n*-GaAs is more than two times larger than those suggested by luminescence measurements[7] and transistor measurements[8]. We are therefore forced to conclude that multiple scattering cannot be neglected in *n*-GaAs because of the importance of low doping concentrations in this semiconductor. However, it appears that even at $N_D > 10^{19} \text{ cm}^{-3}$, multiple scattering is significant in *n*-GaAs. This assertion is supported by the fact that there is considerable distortion

of density of states at high doping in *n*-GaAs[7]. The high-density theories of Berggren *et al.*[11], Sernelius[6,12] and Jain and Roulston are valid when the band shifts as a whole nearly as a rigid band, i.e. when distortion of the band is small. Though multiple scattering theory has been implemented for *n*-GaAs by Serre and Ghazali[13], (see this paper for references to earlier papers of the authors on this subject) and by Bennett and Lowney[9], the procedure is complicated and final results are not obtained in the form of a simple expression. We have therefore numerically fitted a simple expression to the impurity scattering part of a BGN obtained by using multiple scattering theory. We have taken the many-body part from eqn (1). The expression obtained for BGN in this manner and its comparison with experimental results is given in the next section.

3. SIMPLE EQUATION FOR BGN IN *n*-GaAs

3.1. Semi-empirical expression for BGN in *n*-GaAs

The many-body part of BGN, designated as BGN_{mb} , can be described by the first two terms of eqn (1), and by using numerical values of the constants† for *n*-GaAs[1] the many-body part can be written as:

$$BGN_{mb} = 16.30 \left(\frac{N_D}{10^{18}} \right)^{1/3} + 7.47 \left(\frac{N_D}{10^{18}} \right)^{1/4} \text{ meV.} \quad (2)$$

BGN_{mb} values calculated using eqn (2) and those obtained by other workers are shown in Fig. 1. All results agree with each other extremely well. The values of Sernelius[6] are indistinguishable from those calculated using eqn (2) and are not shown in the figure.

BGN_{imp} obtained by multiple scattering[9] theory can be fitted by:

$$BGN_{imp} = 45.7 \left(\frac{N_D}{10^{18}} \right)^{1/3} \text{ meV.} \quad (3)$$

Equation (3) is valid in the doping range $1 \times 10^{18} - 5 \times 10^{19} \text{ cm}^{-3}$.

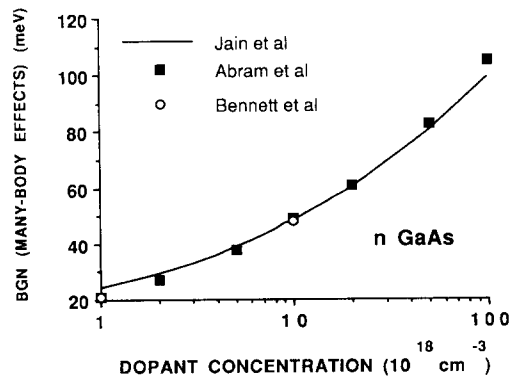


Fig. 1. Many-body bandgap narrowing in *n*-GaAs as a function of doping concentration. Solid curve shows values calculated using eqn (2), symbols show values calculated by other workers[9,10].

†These constants have known values for each semiconductor, they are not fitting parameters. Their values as well as eqn (2) are given in the paper of Jain and Roulston[1].

Combining this with eqn (2), we obtain the following expression for total BGN:

$$\text{BGN} = 62.0 \left(\frac{N_D}{10^{18}} \right)^{1/3} + 7.47 \left(\frac{N_D}{10^{18}} \right)^{1/4} \text{ meV.} \quad (4)$$

The temperature effect on bandgap narrowing is very small and, as discussed by Jain and Roulston[1], equations derived for low temperatures can also be used at room temperature to a good approximation at the doping concentrations considered here. While using eqn (4) to interpret experiments in the next section, we do not use any fitting parameters. As mentioned before, no fitting parameters are required in using eqn (4) for interpreting experiments.

3.2. Comparison with experimental results

BGN calculated using eqn (4) is shown by a solid line in Fig. 2. Experimental results of Yao and Compaan[7] are also shown. The agreement is satisfactory.

The separation between the Γ - and L -valleys in GaAs is only 300 meV[7]. Based on their result that calculated luminescence spectra (neglecting filling of the Γ -valley) agree with the observed spectra, Yao and Compaan[7] suggested that considerable stretching between Γ - and L -bands occurs. A simple calculation shows that for $N_D > 10^{19} \text{ cm}^{-3}$, the Fermi level measured from the conduction band minimum is more than 300 meV, if band distortion does not take place, the upper valley should start filling at higher dopings. Since this has not been taken into account in the BGN theory and since experimental results still agree with the theoretical results, our results support the suggestion of Yao *et al.*[7] that considerable stretching between the Γ - and L -bands occurs in the heavily doped n -GaAs.

Transistor data yield the values of the p - n product and the apparent BGN, designated as ABGN. ABGN is given by[1]:

$$\text{ABGN} = \Delta E_g^a = \text{BGN} - \Delta E^{\text{FD}}, \quad (5)$$

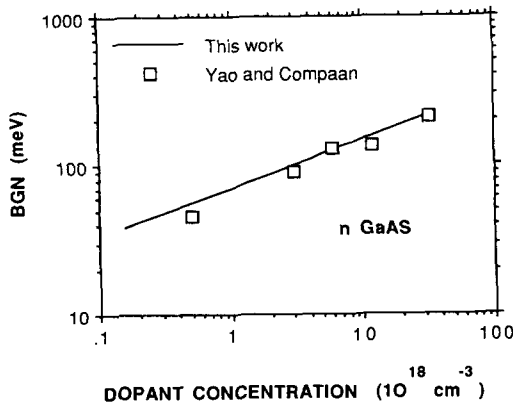


Fig. 2. Bandgap narrowing in n -GaAs as a function of doping concentration. Solid curve shows values calculated using eqn (4), symbols show the luminescence data of Yao and Compaan[7].

Table 1. Minority hole mobility in GaAs used in the present calculations

$N_D (10^{17} \text{ cm}^{-3})$	$\mu_p (\text{cm}^2 \text{ Vs}^{-1})$
1.55	125
6.75	70
11.85	50
24.00	40

where

$$\Delta E^{\text{FD}} = kT \ln \left\{ \exp \left(\frac{-E_F}{kT} \right) F_{1/2} \left(\frac{E_F}{kT} \right) \right\} \quad (6)$$

is the so-called Fermi-Dirac correction. The pn product is given by (see for example Ref. [1])

$$pn = n_{i0}^2 \exp \left(\frac{\Delta E_g^a}{kT} \right) = n_{ic}^2. \quad (7)$$

Harmon *et al.* have plotted experimental values of $(np)D_p$ as a function of doping concentration. To compare the result of eqn (4) with the experimental results of Harmon *et al.*[8], we need the values of the Fermi level E_F and of D_p . Lowney and Bennett[14] have calculated minority hole mobilities in n -GaAs and compared their results with available experimental values. It is seen from their paper that scatter in the experimental values is very large: the actual values used by us taken from their paper are given in Table 1.

Jain and Roulston[1] have shown that in p -type semiconductors the bands are highly distorted on doping heavily and the Fermi level calculated using a low doping density of states does not agree with experimental values. Fermi level E_F of doped crystals can be determined from observed luminescence spectra (see the extensive discussion of this point and relevant references in the paper of Jain and Roulston[1]). We have plotted experimental values of the Fermi level obtained by analyzing[1] luminescence data of earlier workers and compared these with the calculated values (using value of density of states given in the book of Sze[15]) in Fig. 3. It is seen that

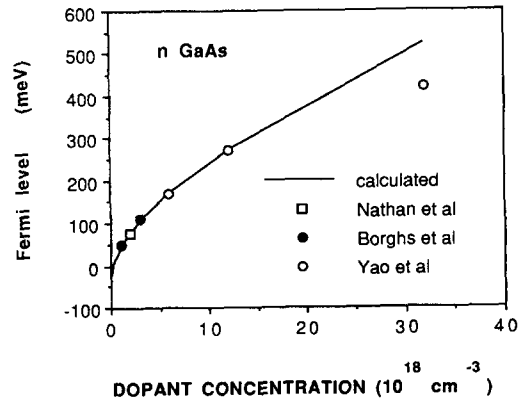


Fig. 3. Fermi level values measured from the conduction band edge in n -GaAs plotted as a function of doping concentration. Solid curve shows the values calculated using density of states value for conduction band in n -GaAs given by Sze[15]. Symbols show the values obtained from the luminescence data published by different authors[7,16,17].

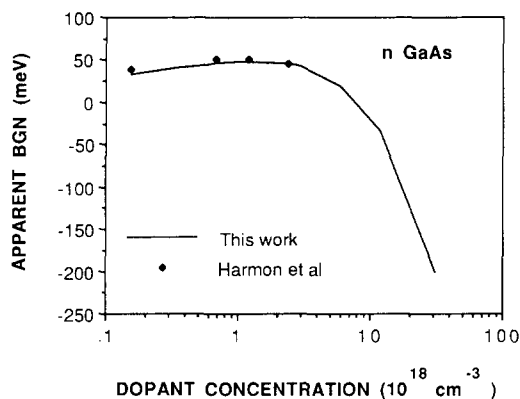


Fig. 4. Apparent bandgap narrowing as a function of doping concentration. Solid line shows the calculated values (see text) and symbols show the experimental values of Harmon *et al.*[8].

at low dopings, agreement between observed and calculated values is good. Above $N_D = 2 \times 10^{19} \text{ cm}^{-3}$, the experimental value is significantly lower as compared to the calculated value, suggesting a distortion of the density of states function on doping *n*-GaAs heavily.

We have calculated apparent BGN using calculated values of E_F (solid curve in Fig. 3) and eqns (5) and (6) and plotted the results in Fig. 4. Using eqns (5), (6) and (7), mobility values given in Table 1 and experimental data of Harmon *et al.* [8] for the $(np)D_p$ product, we have calculated experimental values of apparent BGN also shown in Fig. 4. The agreement is satisfactory. If experimental values of E_F are used, apparent BGN will be somewhat larger than that shown in the figure for $N_D > 2 \times 10^{19} \text{ cm}^{-3}$. However, in the doping range in which transistor measurements have been made[8], the two Fermi levels agree closely.

In conclusion we have obtained a simple expression [eqn (4)] for BGN in *n*-GaAs which agrees well with

recent luminescence and transistor measurements. The equation is based on results of quantum mechanical theories. Multiple impurity scattering cannot be neglected in *n*-GaAs in the doping range in which measurements have been made.

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